

Q. 1 (Binomial distribution gymnastics). a. **Probabilistic Reasoning**

Let B_k be the outcome of a single Bernoulli experiment with probability p . We saw in class that a Binomial random variable is the sum of independent Bernoulli random variables. That is

$$X := B_1 + B_2 + \cdots + B_n$$

and

$$Y := B_{n+1} + B_{n+2} + \cdots + B_{n+m}.$$

It is easy to realize that

$$X + Y = B_1 + B_2 + \cdots + B_n + B_{n+1} + B_{n+2} + \cdots + B_{n+m}$$

must be binomially distributed with $(n + m, p)$. So

$$\mathbb{P}\{X + Y = k\} = \binom{n + m}{k} p^k q^{n+m-k}.$$

Using a conditioning argument

$$\begin{aligned} \mathbb{P}\{X + Y = k\} &= \sum_{y=0}^k \mathbb{P}\{X + y = k | Y = y\} \mathbb{P}\{Y = y\} \\ &= \sum_{y=0}^k \mathbb{P}\{X + y = k\} \mathbb{P}\{Y = y\} \quad \text{by independence of X and Y} \\ &= \sum_{y=0}^k \mathbb{P}\{X = k - y\} \mathbb{P}\{Y = y\} \\ &= \sum_{y=0}^k \binom{n}{k-y} p^{k-y} q^{n-k+y} \binom{m}{y} p^y q^{m-y} \\ &= \sum_{y=0}^k \binom{n}{k-y} \binom{m}{y} p^k q^{n+m-k} \\ &= \binom{n+m}{k} p^k q^{n+m-k}, \end{aligned}$$

where the last equality is obtained by Vandermonde's identity:

$$\sum_{y=0}^k \binom{n}{k-y} \binom{m}{y} = \binom{n+m}{k}$$

b.

$$\begin{aligned}
\mathbb{P}\{X = Z - 1\} &= \mathbb{P}\{Z = X + 1\} \\
&= \sum_{x=0}^n \mathbb{P}\{Z = x + 1 | X = x\} \cdot \mathbb{P}\{X = x\} \\
&= \sum_{x=0}^n \mathbb{P}\{Z = x + 1\} \cdot \mathbb{P}\{X = x\} \\
&= \sum_{x=0}^n q^x p \binom{n}{x} p^x q^{n-x} \\
&= \sum_{x=0}^n p^{x+1} q^n \binom{n}{x} \\
&= pq^n \sum_{x=0}^n \binom{n}{x} p^x \\
&= pq^n (1 + p)^n,
\end{aligned}$$

where we used the binomial formula in the last equality.

Q. 2 (COVID in class). a. Each student being independent and having the same probability of having COVID, the number of students who have COVID during the first lecture follows a *Binomial*(n, p) distribution, where $n = 140$ and $p = 0.005$. The probability we are looking for is then

$$\binom{n}{n/2} p^{n/2} (1-p)^{n-n/2} = \binom{140}{70} (0.005)^{70} (0.995)^{70}$$

b. Let us consider any week W . The probability that someone has COVID during at least one of the lectures of the week W is one minus the probability that no one has COVID during the three lectures of that week. By independence, the probability that no one has COVID during the three lectures is $(1-p)^{3n} = (0.995)^{420}$. Let p_{success} be $1 - (0.995)^{420}$. The weeks being independent, and the probability of “success” being the same for every week, the first week in which someone has COVID during at least one of the lectures is a Geometric random variable with parameter p_{success} .

Q. 3 (Collecting prizes). a. Assume that we already have k distinct types. For every extra cereal box we purchase, there is $1/2$ probability that it will contain a figure, and the figure is the ones we want with probability $(4-k)/4$. The probability that we actually get the desired figures is thus given by $(4-k)/8$. Now the problem can be equivalently formulated as the following: if we repeatedly flip a biased coin with heads probability given by $(4-k)/8$, on expectation how many times we have to flip the coin to see the first heads? Let us denote this number of flips by Y_k . If we use p_k to denote $(4-k)/8$, then $Y_k \sim \text{Geom}(p_k)$ and its expectation is given by

$$\mathbb{E}[Y_k] = \sum_{i=1}^{\infty} i(1-p_k)^{i-1} p_k = 1/p_k.$$

The expected number of boxed we need to eat until we get a new type of figure is thus given by $8/(4-k)$.

- b. If we consider the process of collecting the figures, we can decompose it into 4 stages: the first stage is that we don't have any figures, the second stage is we have exactly one type of figures, and then the stage that we have exactly two, and finally the stage we have exactly three. The expected time we stay at each of the stages, $\mathbb{E}[Y_k]$ for $k \in [0, 3]$, is in fact given by the calculation of the first question. Thus, if Y is denoting the number of total boxes opened until we have the four types, then $Y = Y_0 + Y_1 + Y_2 + Y_3$ and thus the desired expectation is given by

$$\mathbb{E}[Y] = \sum_{k=0}^3 \mathbb{E}[Y_k] = \frac{8}{4} + \frac{8}{3} + \frac{8}{2} + 8 = \frac{50}{3}.$$

- Q. 4** (Monte Carlo). a. By the strong law of large numbers for random variables (Example 2.16 in the lecture notes),

$$\lim_{N \rightarrow \infty} \hat{Z}_N = \mathbb{E}(Z),$$

with probability one. Hence, the error of Monte Carlo approximation $|\mathbb{E}(Z) - \hat{Z}_N|$ can be made arbitrarily small if we choose the number of samples N large enough. This makes Monte Carlo approximation a good one for large values of N .

- b. Since $Z, Z_1, Z_2, \dots$ have the same distribution, $\mathbb{E}(Z) = \mathbb{E}(Z_k)$ and $\text{Var}(Z) = \text{Var}(Z_k)$ for each k . In particular,

$$\mathbb{E}(\hat{Z}_N) = \frac{1}{N} \sum_{k=1}^N \mathbb{E}(Z_k) = \mathbb{E}(Z).$$

We have

$$\begin{aligned} \text{MSE}(N) &= \mathbb{E}((\hat{Z}_N - \mathbb{E}(Z))^2) \\ &= \text{Var}(\hat{Z}_N) \\ &= \text{Var}\left(\frac{1}{N} \sum_{k=1}^N Z_k\right) \\ &= \frac{1}{N^2} \sum_{k=1}^N \text{Var}(Z_k) \\ &= \frac{1}{N^2} \sum_{k=1}^N \text{Var}(Z) = \frac{\text{Var}(Z)}{N} = \frac{1}{N}. \end{aligned}$$

In the fourth equality of this computation, we use $\text{Var}(\sum_{k=1}^N Z_k) = \sum_{k=1}^N \text{Var}(Z_k)$ since $Z_1, \dots, Z_N$ are independent.

Therefore, the smallest number of samples N in order to have $\text{MSE}(N) = \frac{1}{N} < 0.01$ is $N = 101$.

- Q. 5** (More love triangles). Let V denote the set of all individuals on the site ($|V| = n$). For every choice $\{u, v, w\} \subset V$ of three people, let us denote by $X_{\{u, v, w\}}$ the indicator function of the event that $\{u, v, w\}$ forms a love triangle. Note that

$$\mathbb{P}\{X_{\{u, v, w\}} = 1\} = \mathbb{E}(X_{\{u, v, w\}}) = p^3$$

for every $\{u, v, w\} \subset V$.

The total number of love triangles X is given by

$$X = \sum_{\{u,v,w\} \subset V} X_{\{u,v,w\}}.$$

Notice that the indicator functions $X_{\{u,v,w\}}$ are not independent as two triangles might be sharing an edge. We need to compute

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2.$$

In Homework 3, we already computed that

$$\mathbb{E}(X) = \binom{n}{3} p^3.$$

Note that

$$(1) \quad X^2 = \left(\sum_{\{u,v,w\} \subset V} X_{\{u,v,w\}} \right)^2 = \sum_{\{u,v,w\}, \{a,b,c\} \subset V} X_{\{u,v,w\}} X_{\{a,b,c\}};$$

this sum has $\binom{n}{3} \cdot \binom{n}{3}$ terms. Using linearity of expectation, we need to compute the $\mathbb{E}(X_{\{u,v,w\}} X_{\{a,b,c\}})$ for each pair of triangles and add them up. This expectation depends on how many vertices $\{u, v, w\}$ and $\{a, b, c\}$ have in common.

- If $\{u, v, w\} = \{a, b, c\}$, then $X_{\{u,v,w\}} X_{\{a,b,c\}} = (X_{\{u,v,w\}})^2 = X_{\{u,v,w\}}$. Hence,

$$\mathbb{E}(X_{\{u,v,w\}} X_{\{a,b,c\}}) = \mathbb{E}(X_{\{u,v,w\}}) = p^3.$$

- If $\{u, v, w\}$ and $\{a, b, c\}$ have two vertices in common, then the two triangles have only five edges total because they have a common edge. Note that $X_{\{u,v,w\}} X_{\{a,b,c\}}$ is 1 if and only if all of the five edges are present. Hence,

$$\mathbb{E}(X_{\{u,v,w\}} X_{\{a,b,c\}}) = p^5.$$

- If $\{u, v, w\}$ and $\{a, b, c\}$ have less than two vertices in common, then we are concerned about six edges constructed independently. Hence,

$$\mathbb{E}(X_{\{u,v,w\}} X_{\{a,b,c\}}) = \mathbb{E}(X_{\{u,v,w\}}) \mathbb{E}(X_{\{a,b,c\}}) = p^6.$$

To add up the results in the previous paragraph, we need to know how often each occurs in the sum in (1).

- To choose a pair of triangles with $\{u, v, w\} = \{a, b, c\}$, we are only free to choose u, v, w and this can be done in $\binom{n}{3}$ ways.
- To choose a pair $(\{u, v, w\}, \{a, b, c\})$ having two vertices in common: we can choose the first triangle $\{u, v, w\}$ in $\binom{n}{3}$ ways, then we can choose two of u, v, w to be the common vertices in $\binom{3}{2}$ ways, and finally we can choose the remaining “uncommon” vertex in $\{a, b, c\}$ in $(n - 3)$ ways. Hence, we have

$$\binom{n}{3} \cdot \binom{3}{2} \cdot (n - 3) = 12 \binom{n}{4}$$

such pairs of triangles in the sum.

- For the remaining case, we can simply subtract the above counts from all possible ways of choosing $\{u, v, w\}$ and $\{a, b, c\}$. This gives us

$$\binom{n}{3}^2 - \binom{n}{3} - 12\binom{n}{4}.$$

Therefore,

$$\mathbb{E}(X^2) = \binom{n}{3}p^3 + 12\binom{n}{4}p^5 + \left(\binom{n}{3}^2 - \binom{n}{3} - 12\binom{n}{4}\right)p^6$$

Since $(\mathbb{E}(X))^2 = \binom{n}{3}^2 p^6$, we have that

$$\text{Var}(X) = \binom{n}{3}(p^3 - p^6) + 12\binom{n}{4}(p^5 - p^6).$$